EGG DROP

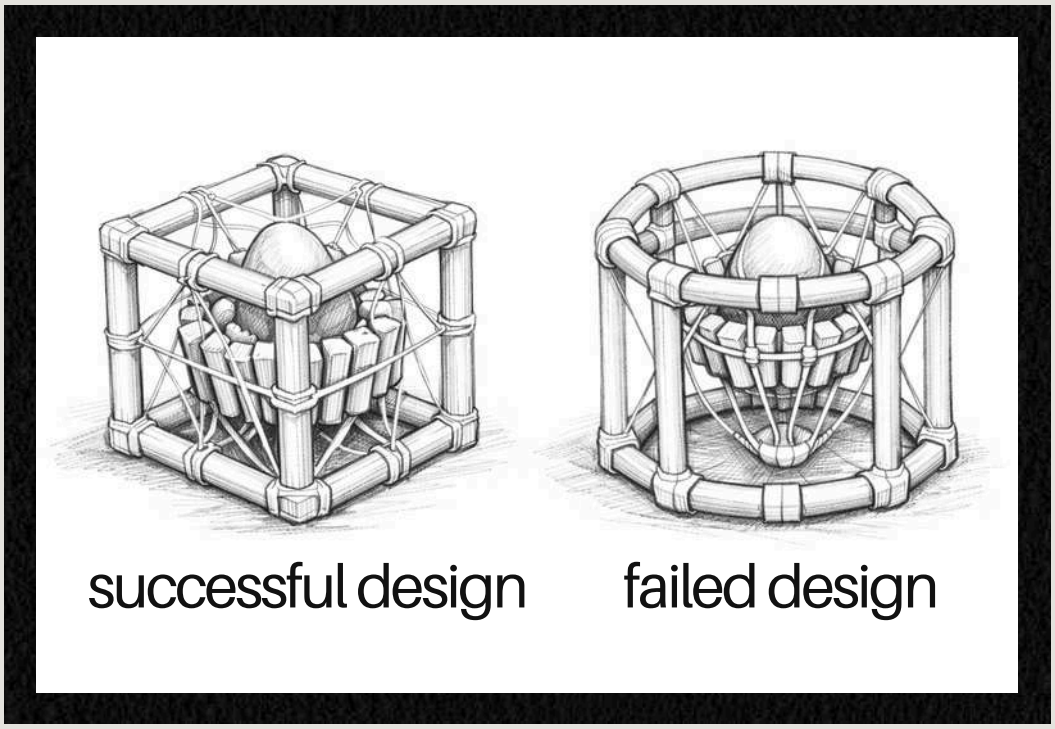
B E S T E
E R B İ L G İ N

13.03.2026

IDE 102

DESIGN MILESTONES

The design process of my egg protection project began with creating a structure to protect an egg from a **three-meter drop**. My first design **was a cube** made with straws and string without strict limits, and it **successfully protected the egg**. Later, the materials were **limited** to 20 straws, 2 meters of string, and 1 meter of tape. **With AI support**, I explored new ideas and chose a **hexagonal prism** design with the egg suspended in the center. Although the design was geometrically strong, **the limited materials made the structure weaker and the egg broke during** the test. This project showed that successful design depends on both geometry and material strength, while AI can help generate ideas but real testing determines the final result.



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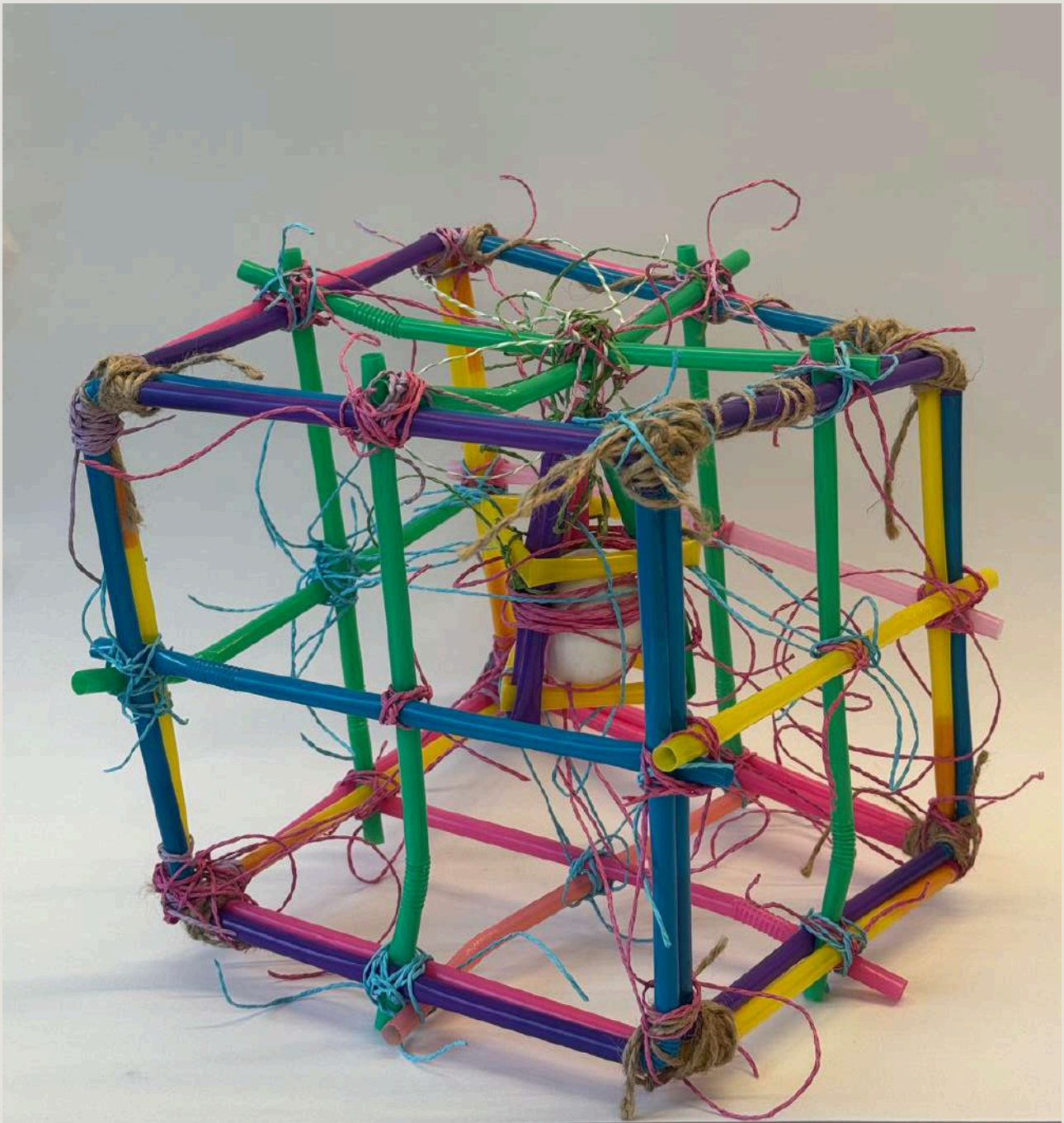
FIRST ATTEMPT

PROCESS

First, we gathered materials such as colorful plastic straws, different types of strings, and a small foam ball. We started by building the outer cube frame by connecting the straws at the corners and tying them tightly with string to keep the structure stable. After completing the frame, we added vertical and diagonal straws inside the cube to strengthen it and create additional geometric forms. Then we placed the foam ball in the center and suspended it by tying strings from different points of the structure so it could stay balanced in the middle. Finally, we added extra strings to reinforce the connections



SECOND ATTEMPT

PROCESS

First, we gathered materials such as colorful plastic straws, tape, and pieces of rope. We began by creating the base of the structure using straws and fixing the corners with tape to form a stable frame. Then we added vertical straws to connect the bottom and top parts, forming a tall geometric shape. After building the outer frame, we placed a small cup-like object in the center and supported it with additional straws and ropes. Finally, we tied ropes between different points of the structure to hold the center piece in place and strengthen the model

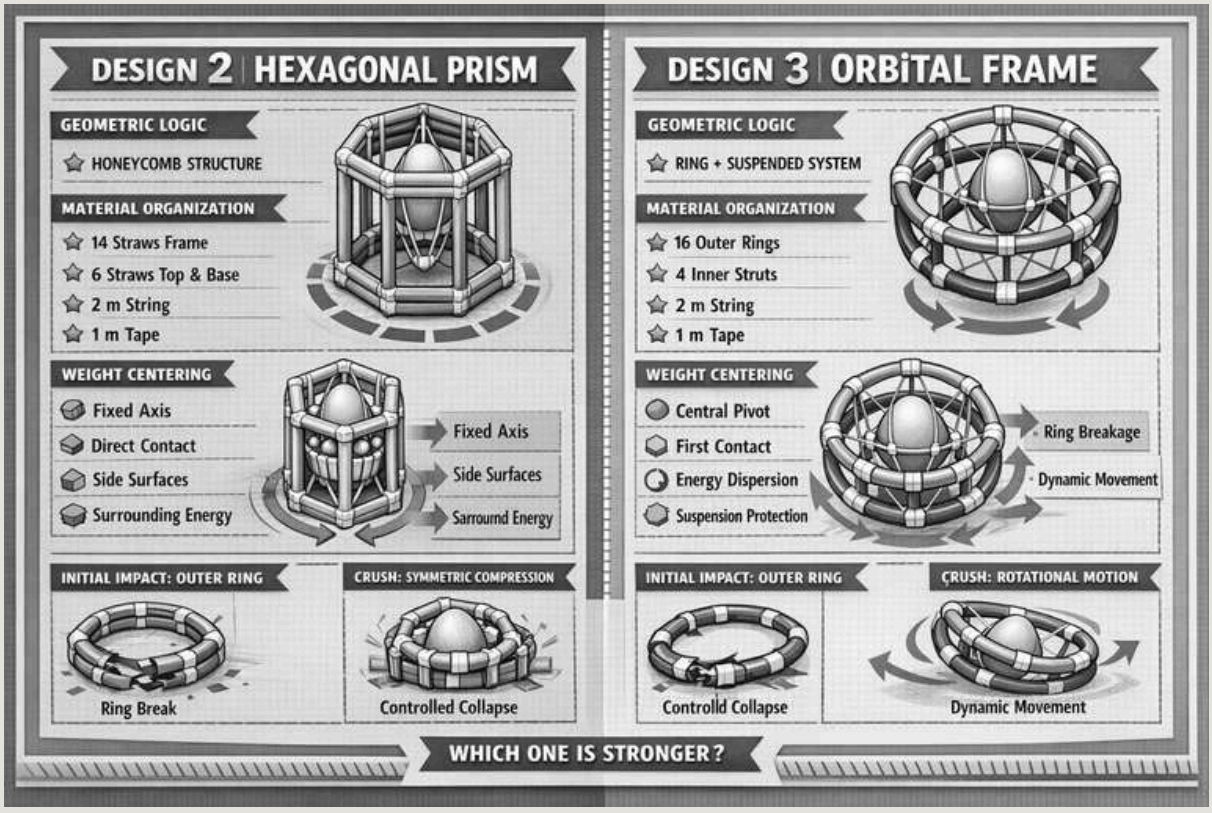
AI PROMPTS

- **Prompt 1:**
  "Explain how structures can protect a fragile object like an egg when dropped from a height of three meters."

- **Prompt 2:**
  "Design a structure that protects an egg using only 20 straws, 2 meters of string, and 1 meter of tape."

- **Prompt 3**
  " Which geometric shape is more durable?"

AI RESPONSE



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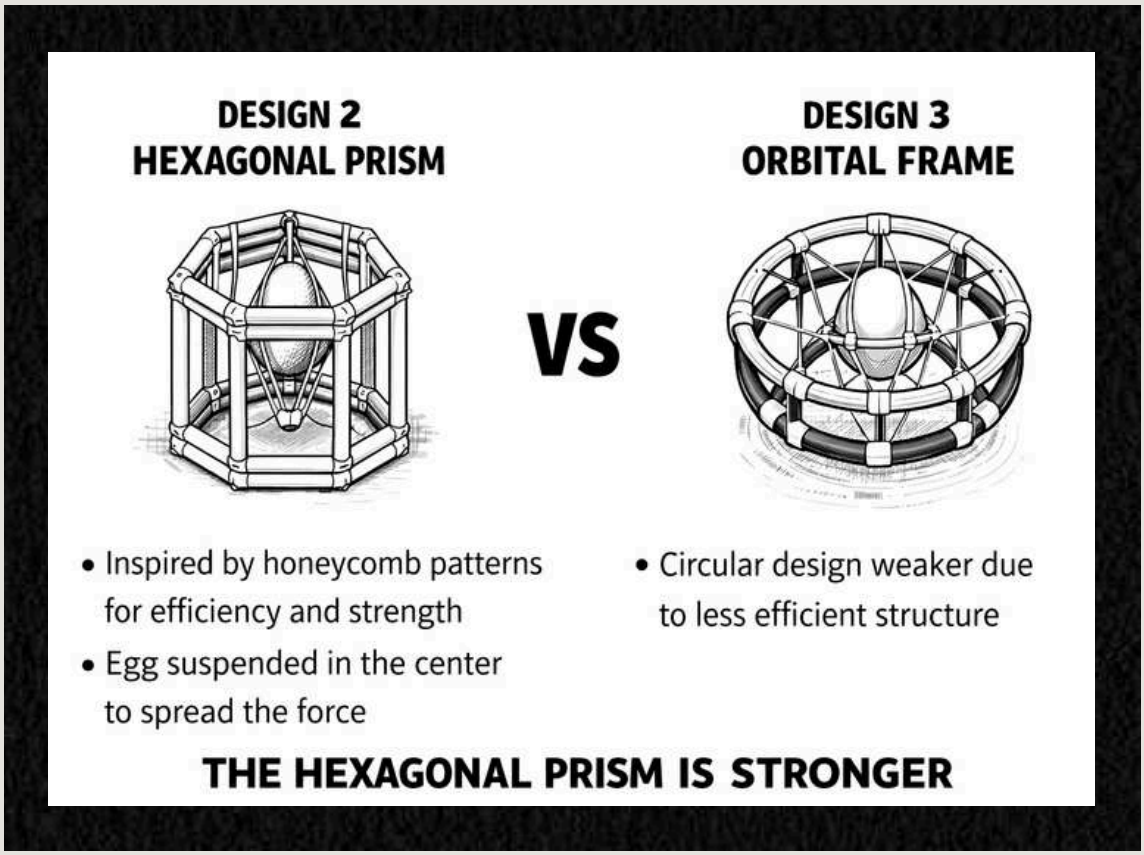
After the AI response I ask several questions

- Which one is stronger? Explain with your reasons.
- Which one would be easier to make with limited materials?



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AI Role Tags

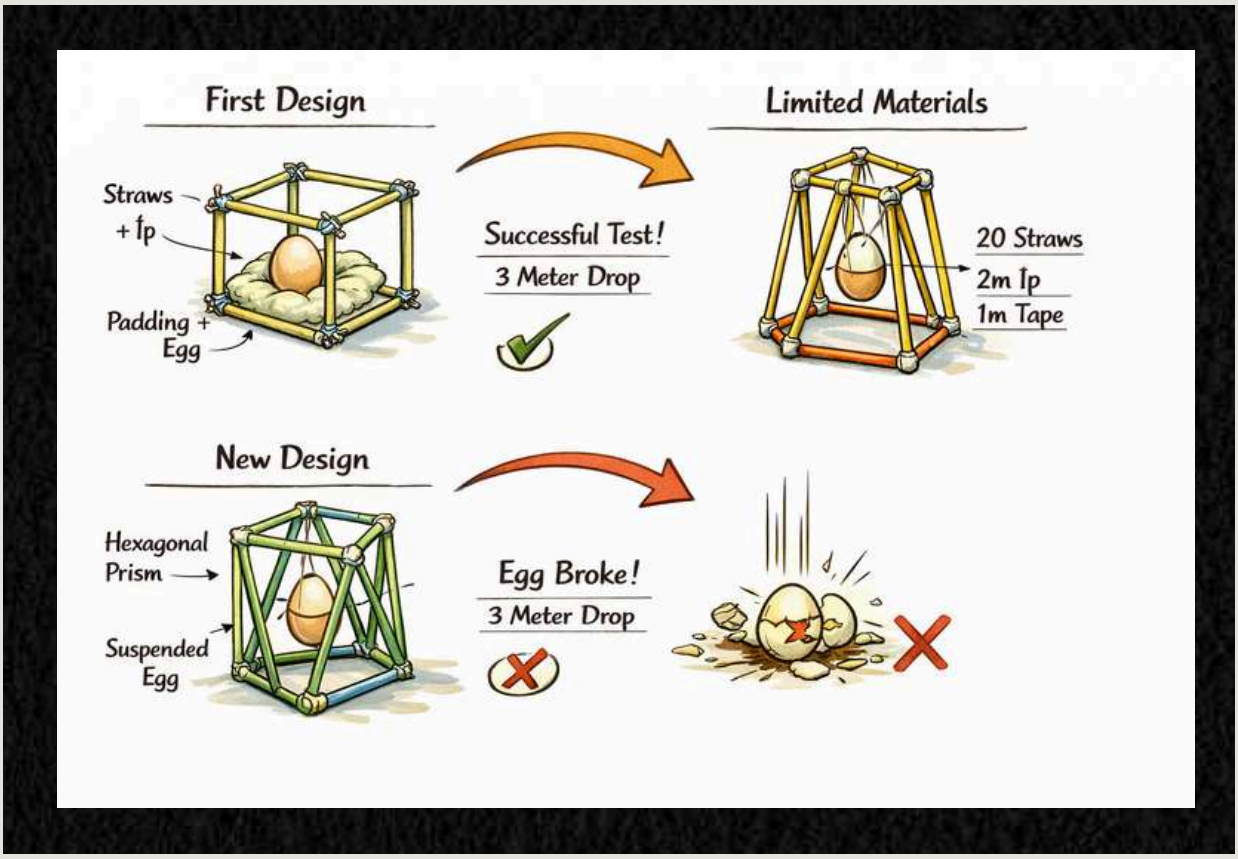
- **Clarifier:** AI helped explain the problem and the importance of impact distribution.
- **Re-framer:** AI suggested looking at the problem through geometric efficiency.
- **Expander:** AI generated multiple structural ideas.
- **Critique:** AI discussed possible weaknesses of each design.

DECISION

The AI suggested several geometric solutions, including circular frames and hexagonal structures. One of the most interesting ideas was a **hexagonal prism** inspired by **honeycomb patterns**, which are known for their efficiency and strength. The hexagonal shape helps distribute force across multiple sides and **creates a balanced and stable frame.** Based on this idea, I built a hexagonal prism structure. The **egg was placed in the center and supported with strings that acted as a suspension system.** This design aimed to reduce direct impact and spread the energy through the frame during the drop.

REFLECTION

Through this project, I learned several important lessons about design:
Strong structures **require a balance between geometry and materials**
Material limitations change the way designers must think about structures
AI can support creativity and idea generation
Real-world testing is necessary to evaluate a design
This process helped me better understand how designers **move from ideas to prototypes and finally to evaluation and reflection.**



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